

BASIS FOR DECISION MEMO

Processor: Mike Bitondo
Permit No. MI0020257

Designated Name: **Cadillac WWTP**

Date: May 22, 2013
PID #: 95475.1

Monitoring Point 001 : Authorization to discharge 3.2 MGD of treated municipal wastewater from Monitoring Point 001A through Outfall 001. Outfall 001 discharges to Clam River.

<u>Parameter</u>	<u>Maximum Limits for Quantity or Loading</u>			<u>Units</u>	<u>Maximum Limits for Quality or Concentration</u>			<u>Units</u>	<u>Frequency of Analysis</u>	<u>Sample Type</u>	<u>Basis for Limits</u>
	<u>Monthly</u>	<u>7-Day</u>	<u>Daily</u>		<u>Monthly</u>	<u>7-Day</u>	<u>Daily</u>				
Flow	(report)	---	(report)	MGD	---	---	---	---	Daily	Report Total Daily Flow	PWJ
Carbonaceous Biochemical Oxygen Demand (CBOD5)											
May 1 – Nov. 30	190	270	---	lbs/day	7	---	10	mg/l	5x/Week	24-Hr Composite	AB
Dec. 1 – Mar. 31	670	910	---	lbs/day	25	---	34	mg/l	5x/Week	24-Hr Composite	AB
Apr. 1 – Apr. 30	450	670	---	lbs/day	17	---	25	mg/l	5x/Week	24-Hr Composite	AB
Total Suspended Solids											
May 1 – Nov. 30	530	800	---	lbs/day	20	30	---	mg/l	5x/Week	24-Hr Composite	PWJ
Dec. 1 – Apr. 30	800	1200	---	lbs/day	30	45	---	mg/l	5x/Week	24-Hr Composite	STS
Ammonia Nitrogen (as N)											
May 1 – Nov. 30	24	53	---	lbs/day	0.9	---	2.0	mg/l	5x/Week	24-Hr Composite	AB
Dec. 1 – Mar. 31	110 99	---	---	lbs/day	4.2 3.7	---	---	mg/l	5x/Week	24-Hr Composite	WQBEL
Apr. 1 – Apr. 30	150 130	---	---	lbs/day	5.5 4.8	---	---	mg/l	5x/Week	24-Hr Composite	WQBEL
Total Phosphorus (as P)											
May 1 – Mar. 31	13	---	---	lbs/day	0.5	---	---	mg/l	5x/Week	24-Hr Composite	WQBEL
Apr. 1 – Apr. 30	17	---	---	lbs/day	0.65	---	---	mg/l	5x/Week	24-Hr Composite	WQBEL
Fecal Coliform Bacteria	---	---	---	---	200	400	---	cts/100ml	5x/Week	Grab	WQS
Acute Toxicity	---	---	---	---	---	---	(report)	TU _A	<i>Monthly</i> Quarterly	24-Hr Composite	PWJ
Chronic Toxicity	---	---	---	---	1.1	---	---	TU _C	<i>Monthly</i> Quarterly	24-Hr Composite	WQBEL
Total Selenium	---	---	---	---	---	---	(report)	µg/l	Quarterly	24-Hr Composite	
Total Zinc	---	---	---	---	---	---	(report)	µg/l	Quarterly	24-Hr Composite	
Available Cyanide	---	---	---	lbs/day	---	---	(report)	µg/l	Quarterly	Grab	

Available Cyanide	0.16	---	---	lbs/day	5.9	---	(report)	µg/l	Weekly	Grab	WQBEL
Total Copper	---	---	---	---	---	---	(report)	µg/l	Quarterly	24-Hr Composite	WQC
Total Mercury	(report)	---	---	lbs/day	(report)	---	---	ng/l	Quarterly	Grab	PWJ
	12-Month Rolling Average				12-Month Rolling Average						
Total Mercury	0.00027	0.000053	---	lbs/day	10 2.0	---	---	ng/l	Quarterly	Calculation	WQV
					Minimum Monthly						
Total Suspended Solids Minimum % Removal Dec 1 – April 30	---	---	---	---	85	---	---	%	Monthly	Calculation	STS
<u>Parameter</u>					<u>Minimum Daily</u>		<u>Maximum Daily</u>				
pH	---	---	---	---	6.5	---	9.0	S.U.	Daily	Grab	WQS
Dissolved Oxygen											
May 1 – Nov. 30	---	---	---	---	6.0	---	---	mg/l	5X/Week	Grab	WQBEL
Dec. 1 – Mar. 31	---	---	---	---	5.0	---	---	mg/l	5X/Week	Grab	AB
Apr. 1 – Apr. 30	---	---	---	---	5.0	---	---	mg/l	5X/Week	Grab	WQBEL

PERMIT CONDITIONS:

Final Effluent Limitations, Monitoring Point 001A
Additional Monitoring Requirements
Pollutant Minimization Program for Total Mercury
Storm Water Pollution Prevention Plan
Untreated or Partially Treated Sewage Discharge Reporting and Testing Requirements
Operation, Maintenance & Replacement/Asset Management
Facility Contact
Monthly Operating Reports
Michigan Industrial Pretreatment Program
Residuals Management Program for Land Application of Biosolids

Limit and Permit Conditions Change Key

Normal Type = existing requirement – carried over from previous version

Bold Type = new requirement – not in previous version

Italic = deleted requirement – not carried over from previous version

Basis for Limits Key

WQBEL - Water Quality Based Effluent Limit

STS – Secondary Treatment Standards

WQS - Water Quality Standards

WQV – Water Quality Variance

PWJ - Permit Writer's Judgment

AB - Antibacksliding